

PAM-SI Update September 2026

SPACIOUS Team

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Devin Johnson, Janelle Gardner,
Kait Palmer, Megan Wood



Kait Palmer
Sound propagation
modeling & simulations

Devin Johnson
Statistical model development

Selene Fregosi
Glider operations & acoustic
data analysis



Megan Wood
Acoustics data analysis &
visualization

Janelle Gardner
Statistical model development

Yvonne Barkley
Acoustic data analysis &
model development

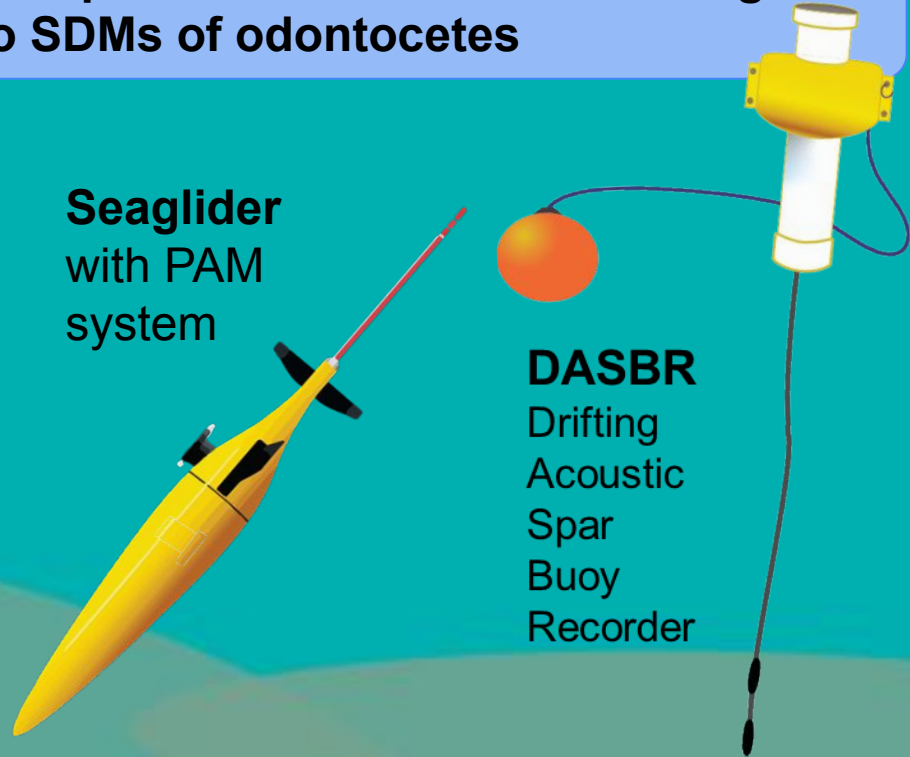
SPACIOUS: Spatial Population Assessment of Cetaceans Integrating acoustic Observations from Uncrewed Systems

Develop analytical framework to incorporate data from slow-moving recording platforms into SDMs of odontocetes

PROJECT GOALS

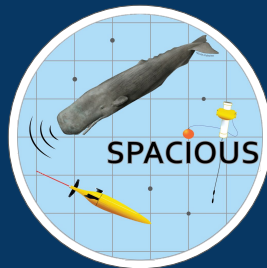
- Identify aspects of spatial models and density estimation to be modified for slow-moving data
- Conduct simulations and experiments to address model components
- Develop density model for sperm whales as case study

Seaglider
with PAM
system



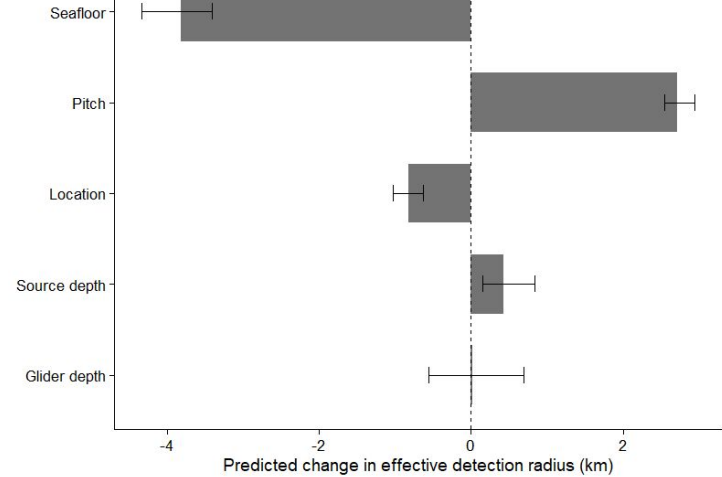
DASBR
Drifting
Acoustic
Spar
Buoy
Recorder

Status Update



TASK SUMMARY

- Finalize propagation modeling sensitivity analysis
- Analysis led by Kait Palmer



Key Accomplishments

- Evaluated the influence of glider depth, environment, and animal behavior on effective detection radius for sperm whale clicks
- For sperm whales, shifts in EDR dominated by animal behavior and environment, not glider depth

Remaining Challenges

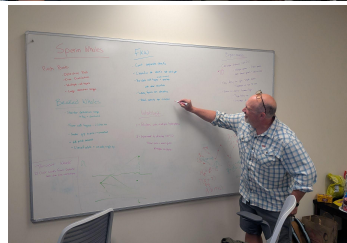
- None

Next Steps

- Finalize draft and submit to journal

TASK SUMMARY

- Develop 'emulator' to approximate detection probability between propagation modelling locations
- Analysis led by Janelle Gardner



**Met in August 2026
for an in-person
workshop focused
on statistical
analyses**

Key Accomplishments

- Created a functional emulator using a GAM to model propagation values between locations

Remaining Challenges

- Improve emulator by running propagation models at more locations

Next Steps

- Test out reduced number of points within propagation models to increase locations (optimize computational efficiency)

TASK SUMMARY

- Complete simulations for likelihood function parameters of a received click
- Analysis led by Devin Johnson



Likelihood for a single received click of level r - developed in Sept 2025 with folks at CREEM

$$\mathcal{L}(r) = \frac{\int f_E(r + L(p)) f_P(p) dp}{\int \int_{u > T+N+L(p)} f_E(u) f_P(p) du dp}$$

Key Accomplishments

- Used the piston model to simulate distribution of attenuation for sperm whale clicks using sound distribution, $s = N(207, 5^2)$

Remaining Challenges

- Determine dive depth distribution
- Fiddle with values to better align effective source level distribution

Next Steps

- Leverage PAMverse R code to calculate sperm whale depth from Towed Array encounters (*Westell et al. 2022*)

TASK SUMMARY

- Establish proof-of-concept modeling framework



Key Accomplishments

- Developed a GAM (encounter rate model) for sperm whales using west coast data from CalCurCEAS 2024
- Best model included SST*Lat, and salinity
- 30% deviance explained!?

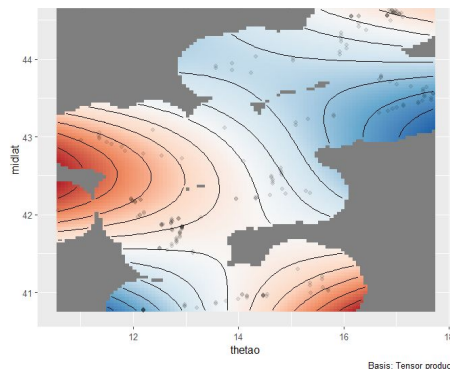
Remaining Challenges

- Incorporate individual clicks per dive segment
- Integrate DASBR data into models

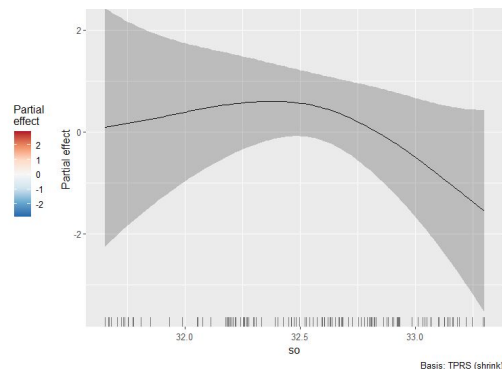
Next Steps

- Validate PAMGuard click detector
- WHICEAS 2026 glider and DASBR data (underway!)
- Pull environmental data for predictions
- Rerun GAM with Tweedie

SST*Latitude

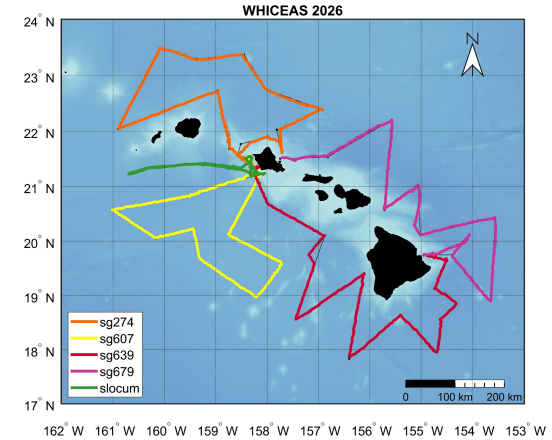
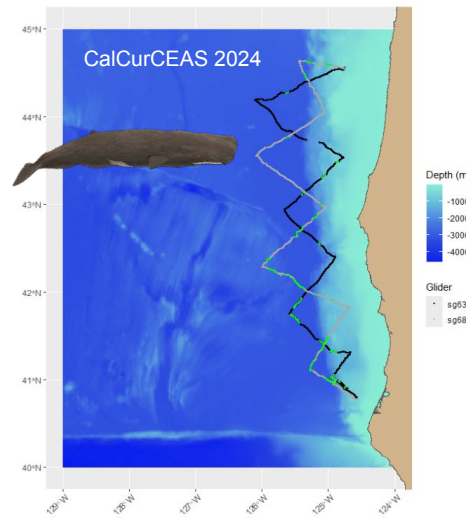


Salinity



TASK SUMMARY

- Data processing and analysis



Key Accomplishments

CalCurCEAS 2024

- Completed manual annotations from 125 encounters, 12 random 1-min subsamples (1500 subsamples) for detector validation

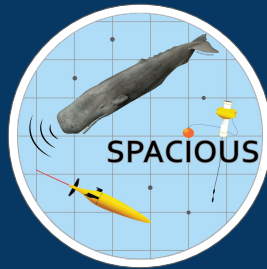
WHICEAS 2026

- 5 gliders deployed
- 4 recorded successfully
- Manual review of recordings underway
- SG274 analysis complete
- SG607 underway

Next Steps

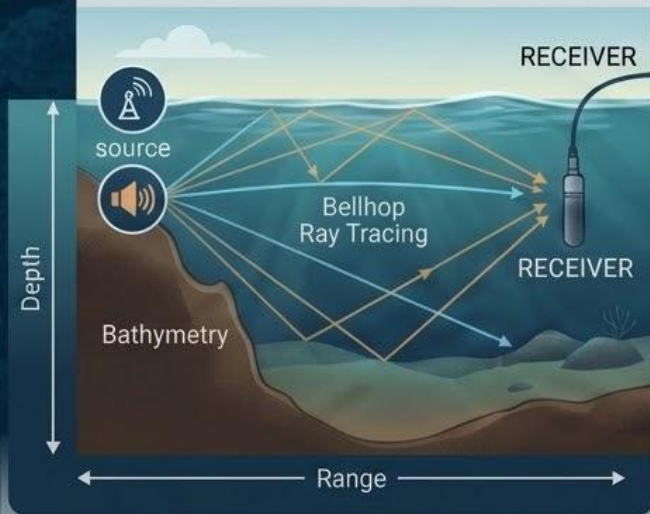
- Validate PAMGuard click detector for CalCurCEAS
- Run detector on WHICEAS data

Products

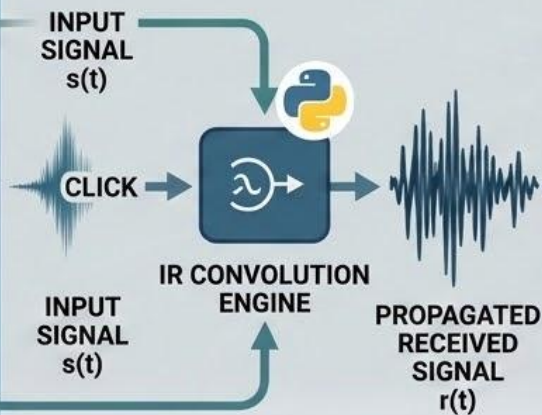


SPATIAL PROPAGATION MODELLING: TWO-STAGE PROCESS ESTABLISHED IN PYTHON

STEP 1: MODEL IMPULSE RESPONSE (IR) WITH BELLHOP

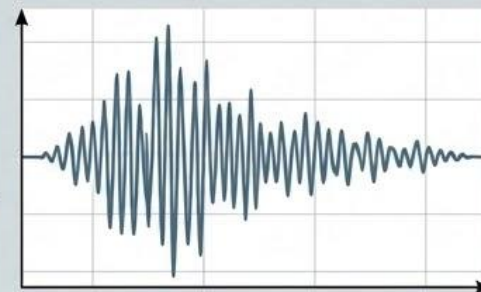


STEP 2: CONVOLVE ACOUSTIC SIGNAL WITH IR



Combine modeled environmental characteristics with any source signal.

STEP 3: EXTRACT & ANALYZE SIGNAL METRICS



- Peak-to-Peak (P2P)
- 10 dB Bandwidth (BW)

Calculate relevant metrics for analysis of detection probability.

Continuous Noise Processing System

YAWN Code Repository for noise processing workflows

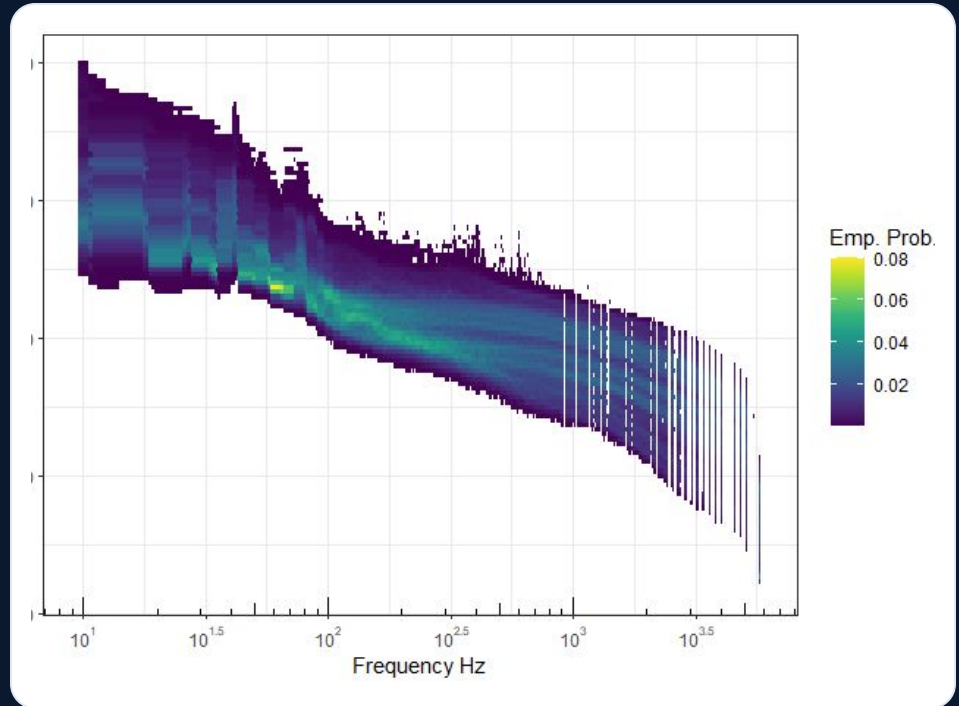
Available on Github:

<https://github.com/PIFSC-Protected-Species-Division/SPACIOUS-NoiseProcessing>

Glider Rodeo Application

Successfully deployed and validated for Glider Rodeo data analysis

- Works across a wide variety of glider acoustic systems
- High-precision spectral density processing



TASK SUMMARY

- 🌟 Track SPACIOUS Progress & Lessons Learned 🌟

 PIFSC-Protected-Species-Division / Projects / PAM-SPACIOUS Retrospective 



 PAM-SPACIOUS Retrospective Add status update Insights Workflows 1  

 Retrospective  Categorize feedback  Action items  New view

 View

 Agenda 🌟 2 

Agenda items for the retrospective

 Draft

Estimating depth distribution from towed array data

 Draft

Determine proportion of clicks have distinguishable p0

 What went well 🟢 1 

What did we do well?

 Draft

Estimate detection area using sound propagation

 What can be improved 🟡 1 

Where can we improve?

 Draft 

Likelihood formulated from effective source level, depth distribution, and spatial distribution

 Action items 🌟 1

Action items so we can improve moving

 Draft

Further simulations of effective source likelihood

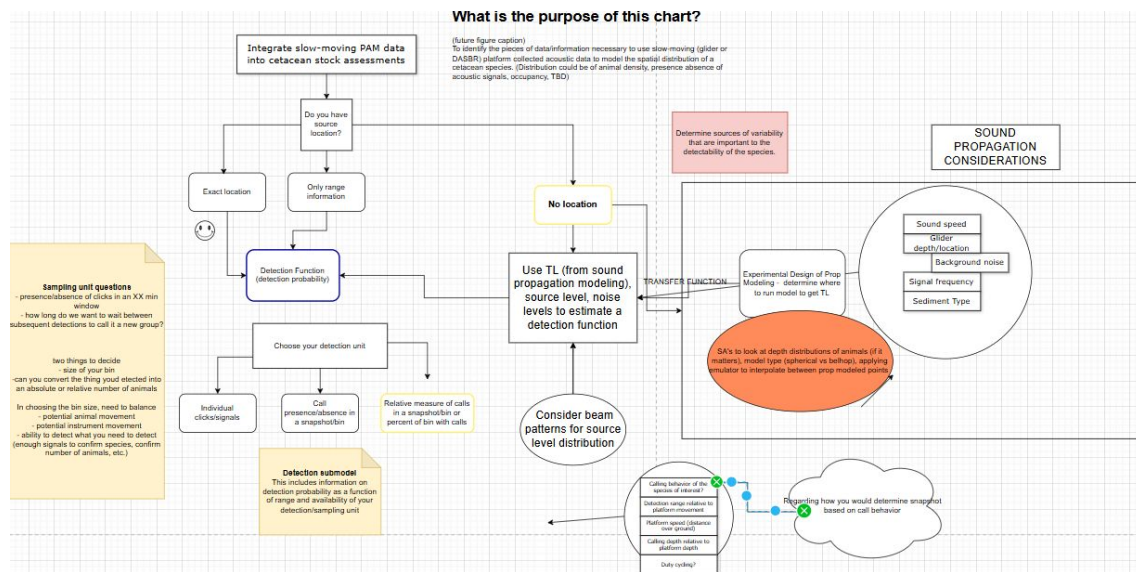
TASK SUMMARY

- Create **flow chart** of the decisions and information required for building spatial models with slow-moving data

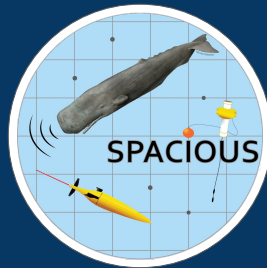
Next Steps

- Complete and share once we finalize the modeling and analysis steps / considerations

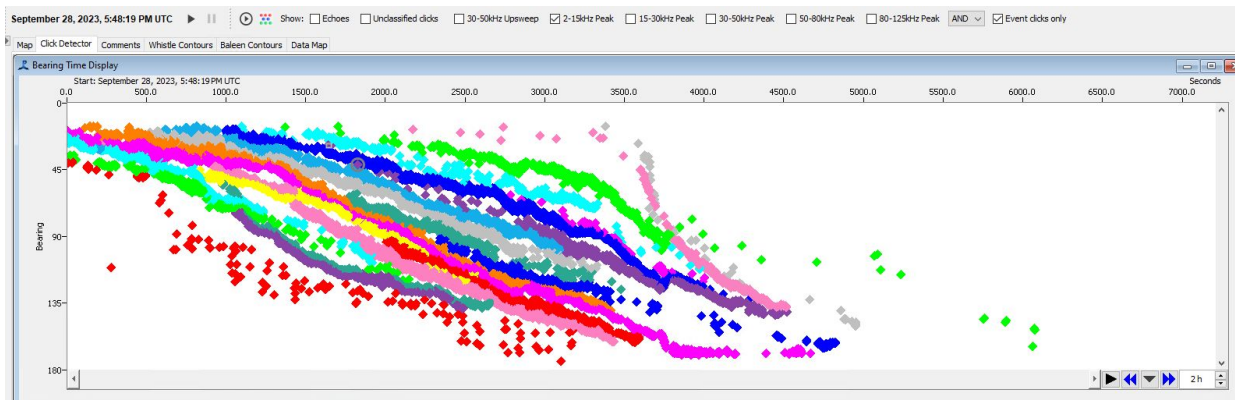
Work In Progress 🧐



Future Ambitions



- Species depth distribution analysis - sperm whales, 2 days to mark



- We didn't forget about DASBRs, we have plans!
- [What species next?](#)
 - beaked whales, false killer whales

Questions?

